

Date
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ENGINEERING MECHANICS

classmate

Date
Page

Mechanics - It is the branch of Engg. Science, deals the effect of forces on any kind of Body which is either in rest or motion.

Classification of Engineering Mechanics :-

There are basically or broadly classified in two parts :-

- ① on the basis of type of Body.
- ② on the basis of static.

1) On the basis of type of Body -

a) Rigid Bodies Mechanics :-

b) Deformable Bodies Mechanics :-

c) fluid Bodies Mechanics :-

Force - Force is something which change or tends to change the state of rest or Uniform motion of a body in a str. line.

classmate
Date _____
Page _____

It is direct or indirect action of one body to another body. The body may be in a direct contact with each other causing direct motion or separated by distance.

Different Types of forces :-

- ① Gravitational force
- ② Frictional force
- ③ Magnetic force

Force is a Vector Quantity because it has both Magnitude and direction.

When a force act on a body the following effect may be produced in that body -

- ① It may bring a change in the Motion of Body.
- ② It may balance the forces already acting on a body thus bringing the body to state of rest or Equilibrium.

- ③ It may change the shape, size and direction of the body.

In Engineering Mechanics, we can classify into two parts -

① Static

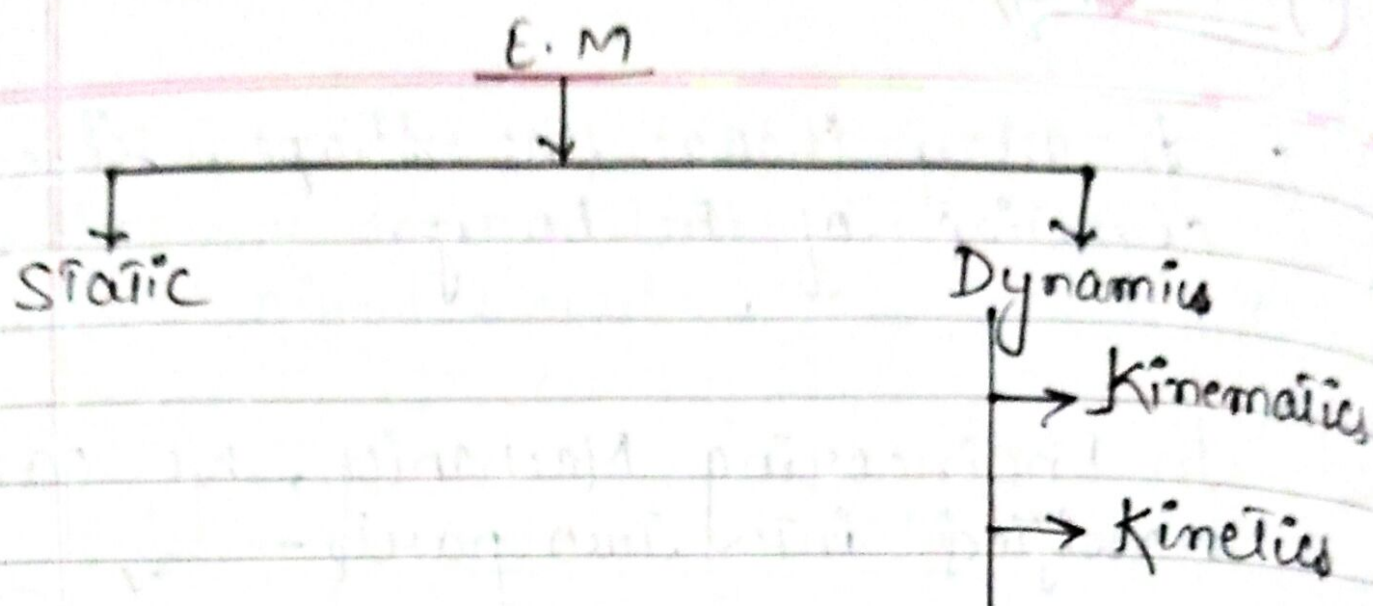
② Dynamic

① Static - Static which relates to the bodies at rest when we apply the external force to the body.

Ex - Sitting on a bench,

② Dynamic - Dynamic which relates to the bodies at motion when we apply the external force on it.

Ex - Kick on a football.



Kinematics → It deals the motion of bodies without any reference to the cause of motion.

Kinetics → It deals the motion of bodies with any reference to the cause of motion.

~~Applied~~ $\neq$

Absolute Unit → Because the mass and acceleration are measured differently in different system of unit so that the unit of forces are also in different in various system.

① FPS system - (Foot Pound Second)

② C.G.S. system - (centimetre gram system)

③ M.K.S. system - (Metre Kg second)

S.I. based Unit

Quantity	Name	Symbol
→ Length	Meter	m
→ Mass	kg (kilogram)	kg
→ Time	Second	s
→ Electric Current	Ampere	A
→ Temperature	Kelvin	K
→ Amount of Substance	Mole	mol
→ Luminous Intensity	Candela	Cd

Second Class of S.I. Derived Unit

Quantity	Name	Symbol
→ Area	m^2	
→ Volume	m^3	
→ Speed / velocity	m/s	
→ Acceleration	m/s^2	
→ Wave number	$\frac{1}{m}$	
→ Density / Mass Density	$\frac{M}{V} \Rightarrow \frac{kg}{m^3}$ $\Rightarrow kg m^{-3}$	

S.I. Derived Special Unit

Quantity	Name	Symbol	Expression in Terms of S.I unit
→ frequency	Hertz	(Hz)	s^{-1}
→ Force (Maxwell)	Newton	(N)	$kg\,m/s^2$
→ Pascal → Pressure	Pascal	(P)	$N/m^2 (kg\,m^{-1}\,s^{-2})$
→ Energy	Joule	(J)	$N\,m (kg\,m^2\,s^{-2})$
→ Gradient / Quantity	Watt	(W)	$J/s (kg\,m^2\,s^{-3})$

Force - 1 N is denoted as a $kg\,m/s^2$. (0.012 kgf)

$$1\,kgf \Rightarrow 9.81\,Newton$$

Pressure -

$$1\,bar \Rightarrow 750.06\,mm\,of\,Hg$$

$$Hg \Rightarrow 0.9869\,atm \Rightarrow 10^5\,N/m^2$$

$$(10^3\,kg\,m^{-1}\,s^{-2})$$

Non-coplanar concurrent force $\rightarrow$ The forces, which meet at one point, but their lines of action do not lie on the same plane, are known as non-coplanar concurrent forces.

$$1 \text{ N/m}^2 \Rightarrow 1 \text{ Pascal} \Rightarrow 10^{-5} \text{ bar}$$

$$\Downarrow$$
$$10^{-2} \text{ Kg m}^{-1} \text{ s}^{-2}$$

$$1 \text{ atm} \Rightarrow 760 \text{ mm of Hg} \Rightarrow 1.03 \text{ Kgf/cm}^2$$

$$\Downarrow$$
$$1.01325 \text{ bar}$$

$$\Downarrow$$
$$1.01325 \times 10^5 \text{ N/m}^2$$

Force System

When two or more than two forces are acting or applied to the body which is called as force system.

Types of force system

There are basically three types of force system :-

① Parallel force system

② Concurrent force system

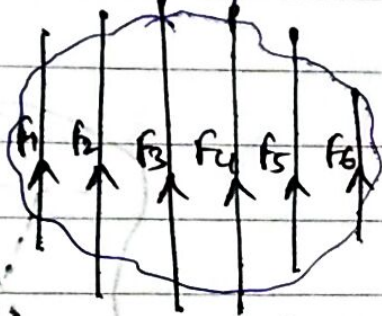
Coplanar Concurrent forces $\rightarrow$ The forces, which meet at one point and their lines of action also lie on the same plane are known as coplanar concurrent forces.

Page _____

③ Collinear force system

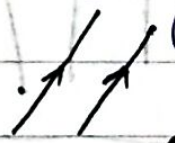
① Parallel force system - If the line of action of all the forces are parallel to each other.

Ex -

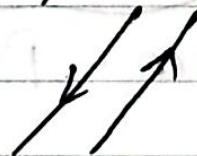


Parallel force system also categorised as two form - force

① Like parallel system -

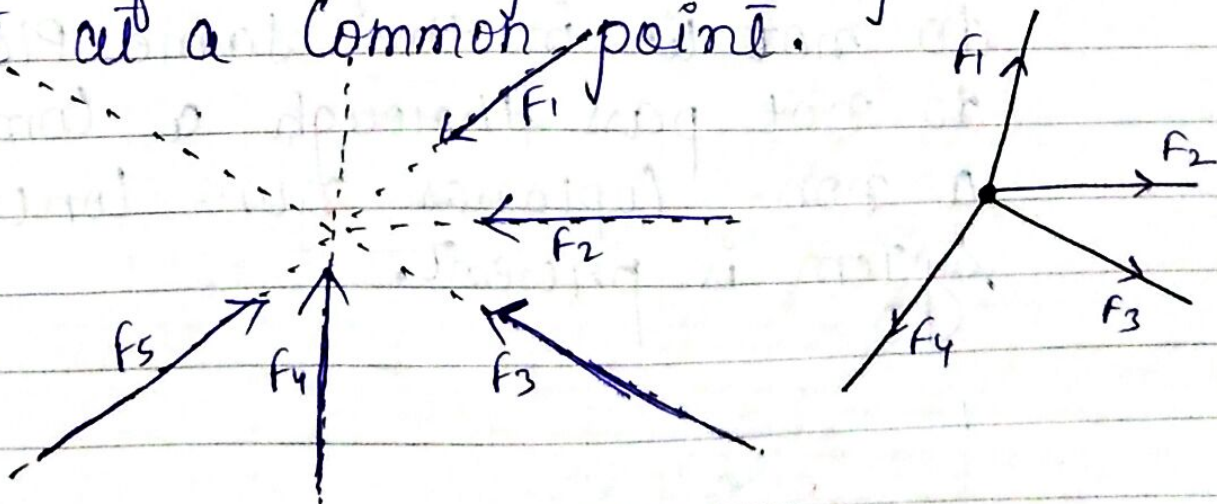


② Unlike Parallel system -



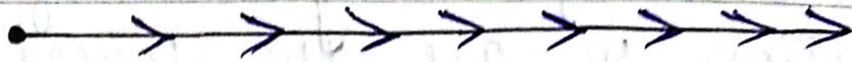
② Concurrent force system - If the line of action of all the force are passes or meet at a common point.

Ex -

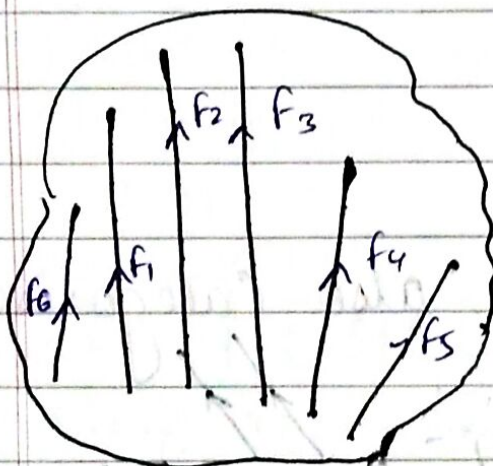


- ③ Collinear force System - If the line of action of all the forces are passes through a single line.

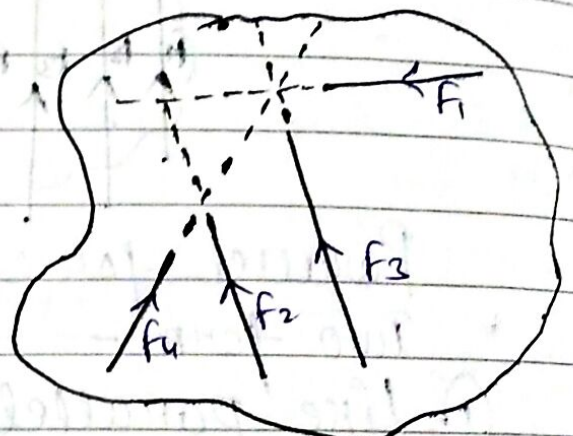
Ex-
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Ex-
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Non - Parallel



Non - Concurrent

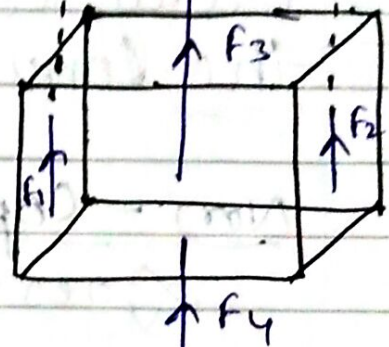
Non - Coplanar Non - Concurrent System
where the lines of action of all forces do not lie in the same plane and do not pass through a common point, a non-coplanar non-concurrent system is present.

2D

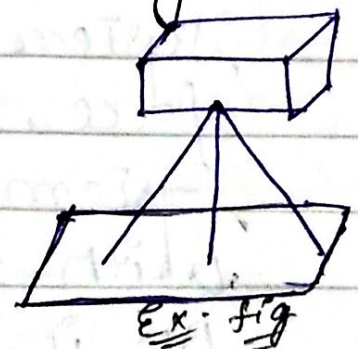
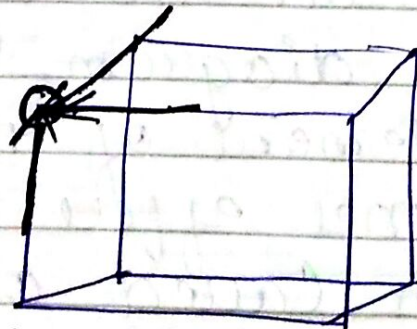
(a) Coplanar Parallel force system

3D

(a) Non-Parallel force system.



Non - Coplanar Concurrent System



The lines of action of all forces do not lie in the same plane but do pass through a common point. An ex. of this force system is the forces in the legs of Tripod support for camera.

classmate
Date _____
Page _____

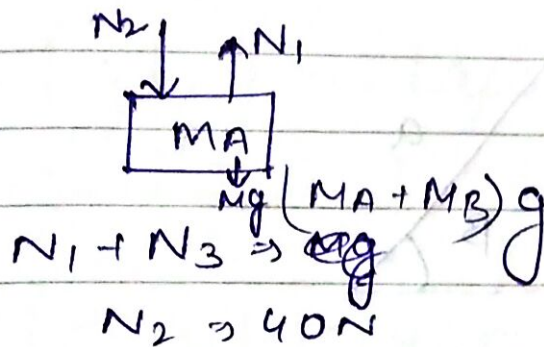
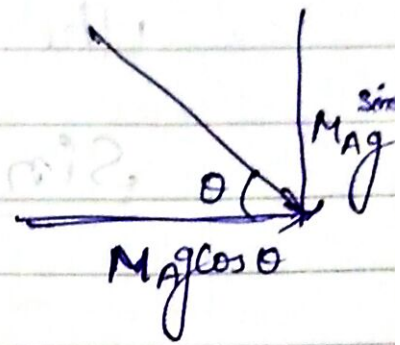
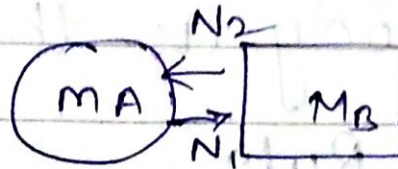
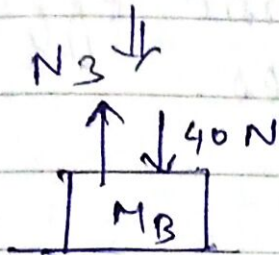
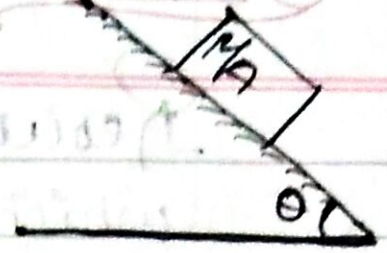
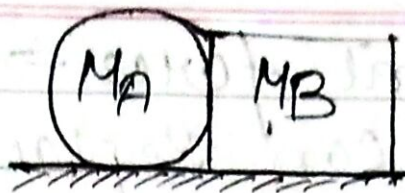
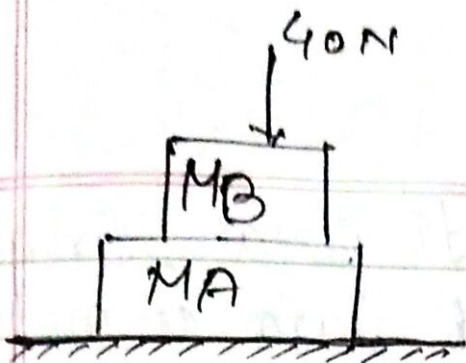
Coplanar Collinear force System

- ① Concurrent force system need not be collinear force system.
- ② Every collinear force system may be the concurrent force system.

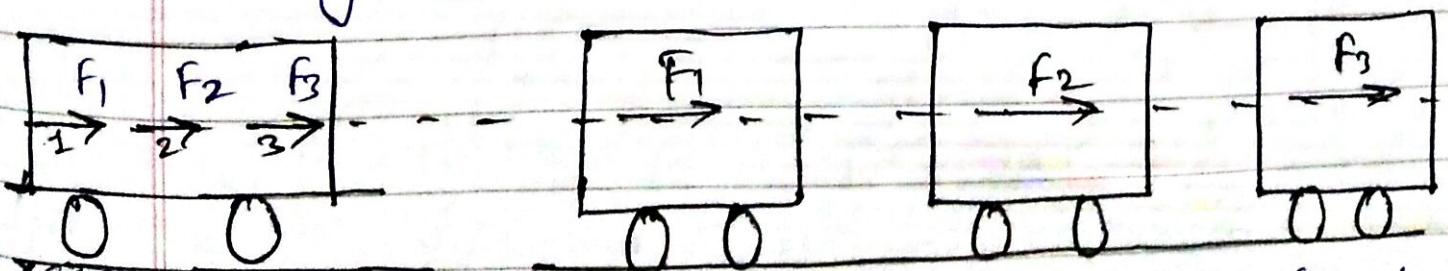
Non-Coplanar Collinear force System
Doesn't exist.

Free Body Diagram:-

A body may consist of more than one element and supports each element and support can be isolated from the rest of the system by incorporating the net effect of the remaining system instead through a set of force. This diagram of the isolated system - element of the body along with the net effect of the system on it, which called as free - Body Diagram.



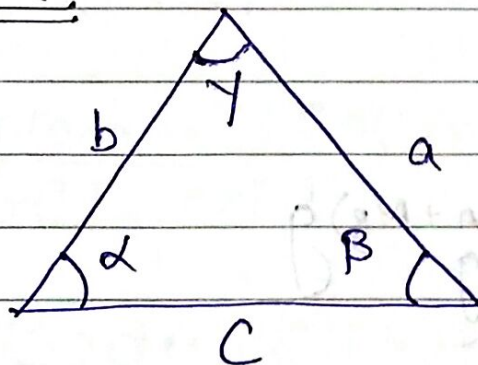
Transmissibility of a force - The principle of Transmissibility of forces state that when a force act upon a body, its effect is the same whatever point in its line of action is taken as the point of the application provided that point is connected with the rest of the body in the same invariable manner.



ex. Train is connected with all his coach

Resultant force $\rightarrow$ It is a single force which can replace two or more forces and produced the same effect on the body as the forces.

Sine Rule



$$\frac{a}{\sin \alpha} = \frac{b}{\sin \beta} = \frac{c}{\sin \gamma}$$

$$b = \sqrt{a^2 + c^2 - 2ac \cos \beta}$$

$$a = \sqrt{b^2 + c^2 - 2bc \cos \alpha}$$

$$c = \sqrt{a^2 + b^2 - 2ab \cos \gamma}$$

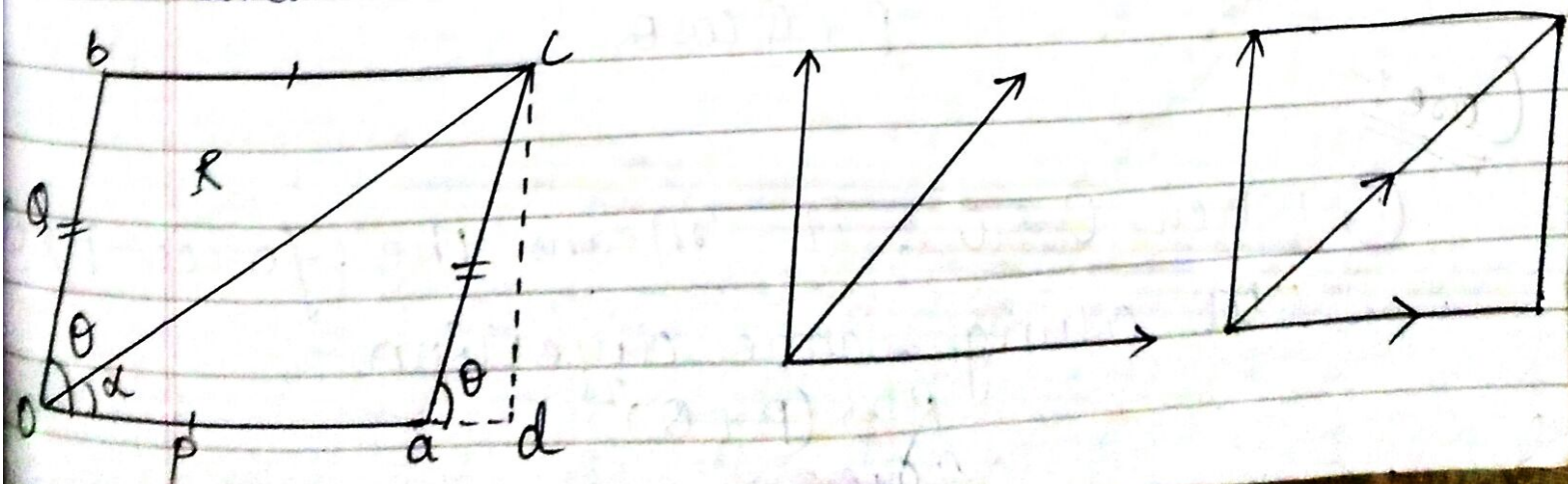
Law of forces

The Various which is used for the Composition of forces are as follows:-

- ① Parallelogram law of forces -
- ② Triangle law of forces -
- ③ Polygon law of forces -

1) Parallelogram law of forces -

If two forces acting simultaneously on a Particle be represented in magnitude and direction by the two adjacent side of a Parallelogram then their resultant may be represented in magnitude and direction by the diagonal of parallelogram which passes through the Point of their intersection.



$$L_{cad} \rightarrow L_{bod} = 0$$

ac is represent as 'Q', $cd = Q \sin \theta$
 $ad = Q \cos \theta$

From right angle $\triangle odc$,

$$OC = \sqrt{(od)^2 + (cd)^2}$$

$$\Rightarrow \sqrt{(P + Q \cos \theta)^2 + Q^2 \sin^2 \theta}$$

$$R = \sqrt{P^2 + Q^2 \cos^2 \theta + 2PQ \cos \theta + Q^2 \sin^2 \theta}$$

$$R = \sqrt{P^2 + Q^2 + 2PQ \cos \theta}$$

$$\tan \alpha = \frac{cd}{od} = \frac{cd}{oa + ad}$$

$$\tan \alpha = \frac{Q \sin \theta}{P + Q \cos \theta}$$

Case:-

① when $\theta = 0^\circ$, it means the force P & Q
 out along same direction -
 $R = (P + Q)^2$
 $\tan \alpha = 0$

② If $\theta \Rightarrow 90^\circ$

$$R \Rightarrow \sqrt{P^2 + Q^2}$$

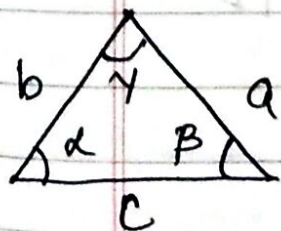
③ If $\theta \Rightarrow 180^\circ$

$$\cos \theta \Rightarrow -1$$

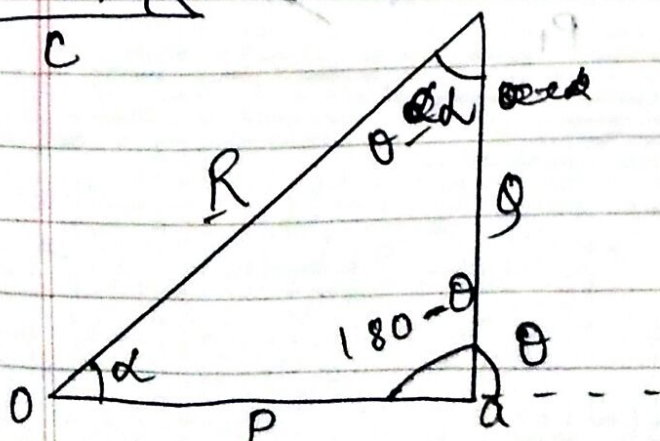
$$R \Rightarrow P - Q$$

27 Triangle law of forces -

If Two forces acting simultaneously on a body are represented in Magnitude and direction only by the Two sides of Triangle taken in order, then their Resultant may be represent in Magnitude and direction by the third side taken in opposite order.



$$\frac{a}{\sin \alpha} \Rightarrow \frac{b}{\sin \beta} \Rightarrow \frac{c}{\sin \gamma}$$



$$\angle COA \Rightarrow \alpha$$

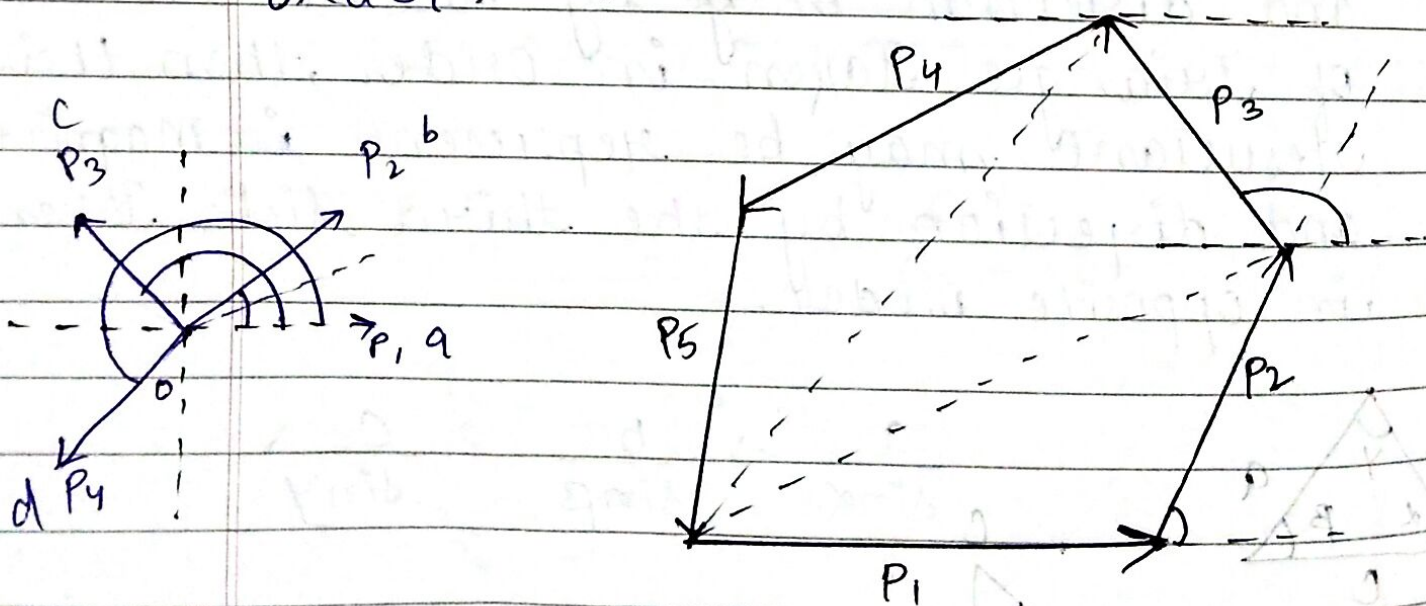
$$\angle OCA \Rightarrow \theta - \alpha$$

$$\angle OAC \Rightarrow 180 - \theta$$

$$\frac{R}{\sin(180 - \theta)} \Rightarrow \frac{P}{\sin(\alpha - \theta)} \Rightarrow \frac{Q}{\sin \alpha}$$

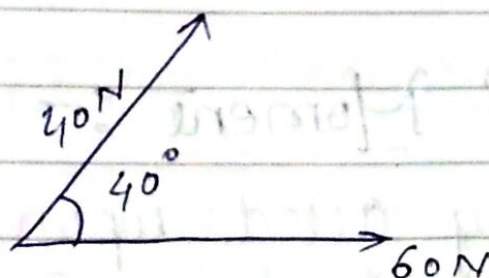
3) Polygon law of forces -

If a number of coplanar concurrent forces acting simultaneously on a body are represented its magnitude and direction by the side of Polygon taken in Order then their Resultant may be represented its magnitude and direction by the closing side of a Polygon taken in the Opposite order.



Q Find the magnitude & direction of the resultant of two forces 40 N & 60 N acting at a point with an included angle of 40° between them. The force of 60 N being horizontal.

Sol.

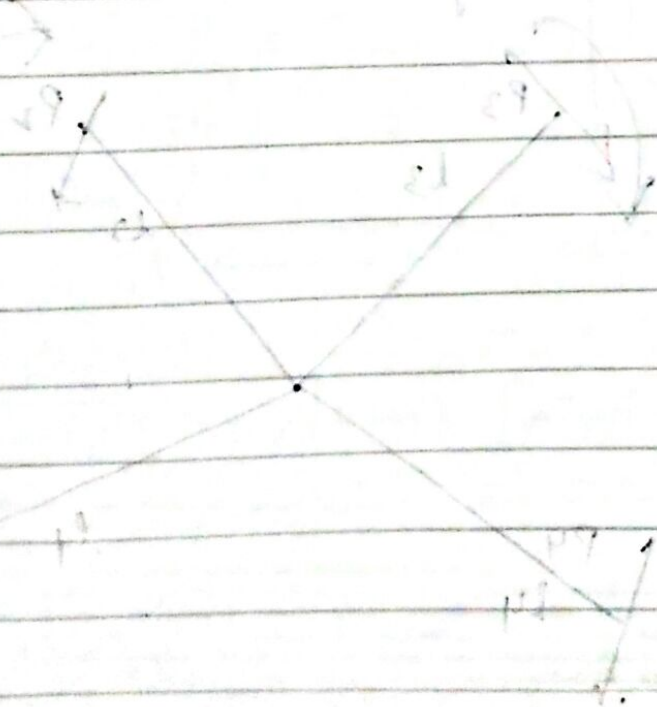


$$\tan \alpha = \frac{P \sin \theta}{P \cos \theta + Q}$$

$$R = \sqrt{P^2 + Q^2 + 2PQ \cos \theta}$$

$$\tan \alpha = \frac{40 \sin 40^\circ}{60 + 40 \cos 40^\circ} \Rightarrow \frac{40 \times 0.74}{60 + 40 \times (-0.66)}$$

$$R = \sqrt{3600 + 1600 + 4800 \cos 40^\circ}$$



P	10 (+)	10
P	10 (-)	10
P	10 (-)	10
P	10 (+)	10

Moment

The tendency of forces is not only to move the body but also rotate the body. The rotational tendency of a force is called as Moment.

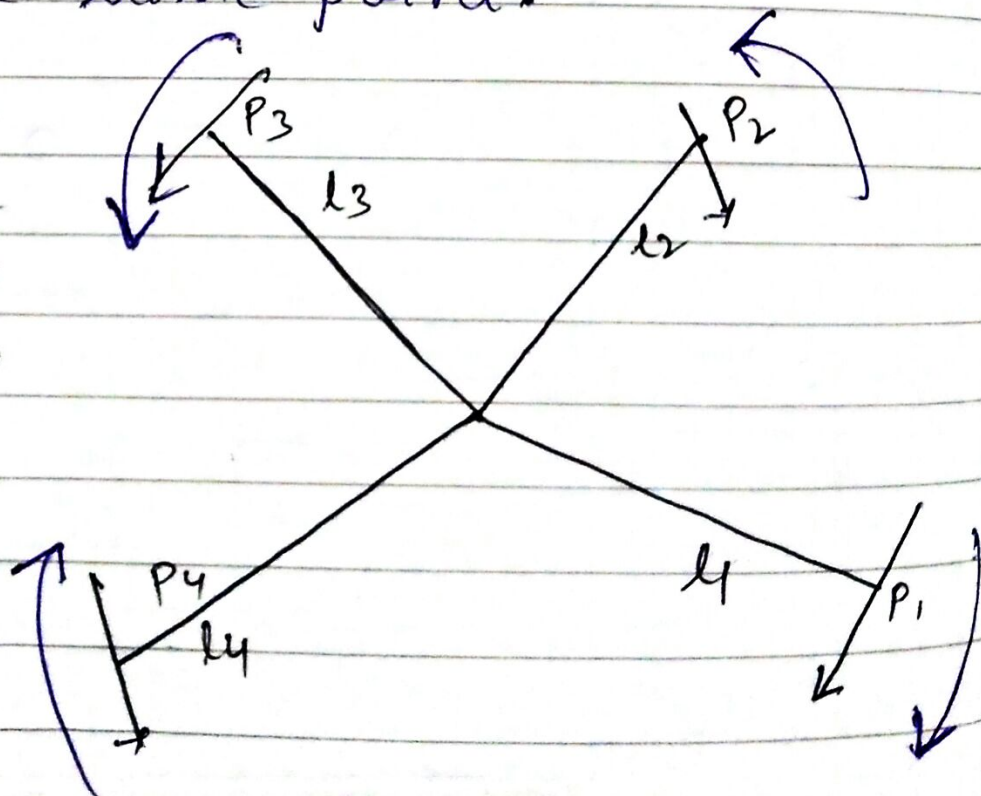
$$\text{Moment} = \text{Force} \times \perp^{\text{r}} \text{dis.}$$

Principle of Moment :-

When a body acted upon by a several forces is in rotational equilibrium, the sum of the clockwise moment of the forces about any point is equal to the sum of the anticlockwise moment of the forces about the same point.

Ex-

⊕	P_1	L_1
⊖	P_2	L_2
⊖	P_3	L_3
⊕	P_4	L_4

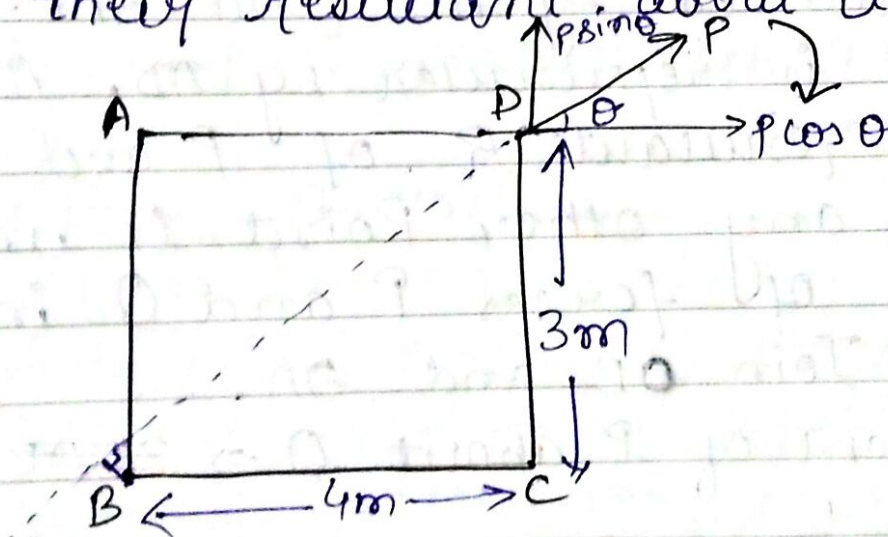


The resultant moment i.e. the algebraic sum of all the moment.

$$\text{Resultant} \Rightarrow P_1 L_1 - P_2 L_2 - P_3 L_3 + P_4 L_4$$

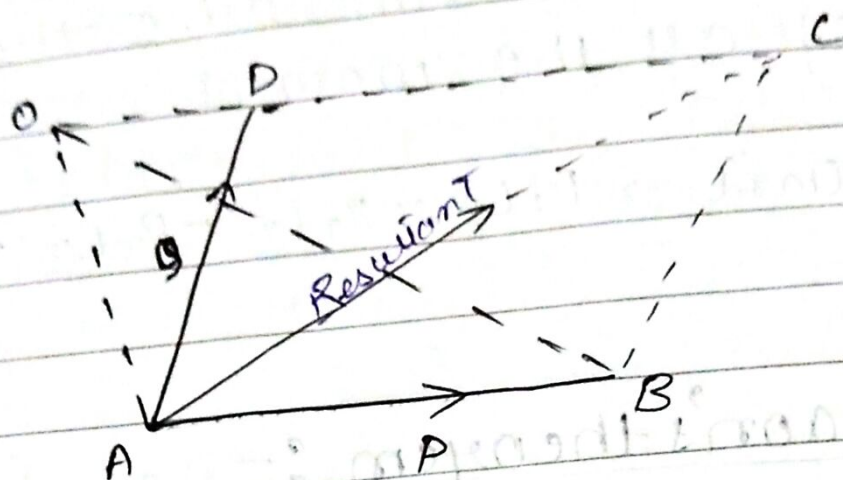
Varignon's theorem :-

It states that the algebraic sum of moment of two forces about any point in their plane is equal to moment of their resultant about that point.



$$\text{Moment at Point B} = (P \cos \theta \times 3) + (P \sin \theta \times 4)$$

Ex.



Case ① when the two forces meet at a point - Two forces P and Q acting at a point A. The magnitude of P is represented by AB and the magnitude of Q is represented by AD. AC represents the resultant R of P and Q.

Take any other point O in the plane of forces P and Q in the line CD. Join OB and OA.

Moment of P about O $\rightarrow$ ~~$2 \times \Delta OAB$~~ $2 \times \Delta OAB$

Moment of Q about O $\Rightarrow 2 \times \Delta OAD$

Moment of Resultant about O $\Rightarrow 2 \times \Delta OAC$

Adding, Algebraically the moment of P and Q - $2(\Delta OAD + \Delta OAB) \Rightarrow \underline{\underline{2 \times \Delta OAC}}$

Couple

A Couple is a pair of two equal and opposite forces acting on a body in such a way in that the line of action of the two forces are not in the same straight line.

- Q The angle between two forces of magnitude 20N & 15N is 60° , the 20N force is horizontal. Determine the resultant of magnitude & direction if the forces are
- ① Pulling condition
 - ② The 15N force is a Pushing condition.
 - ③ and 20N is a pull conditions.

Soln $P \Rightarrow 20\text{N}$, $Q \Rightarrow 15\text{N}$

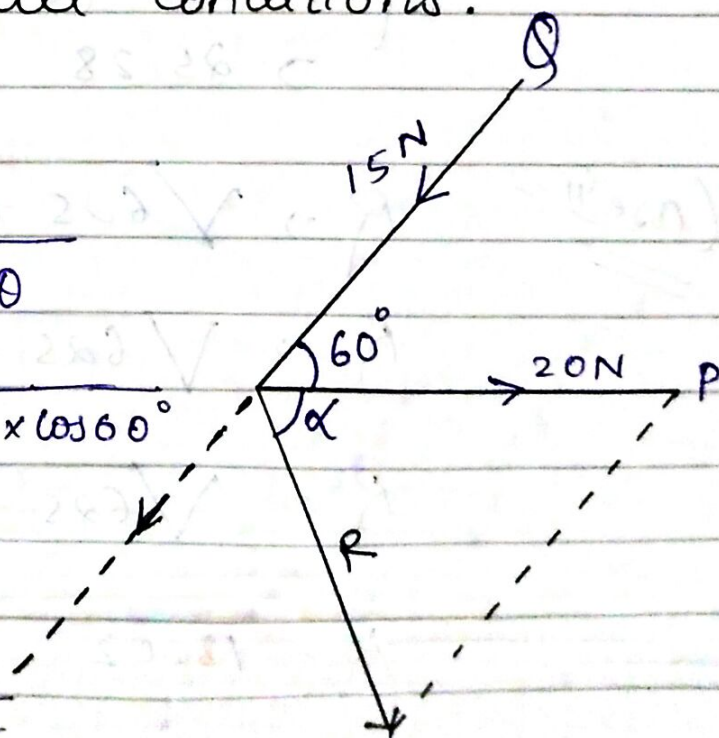
$$R \Rightarrow \sqrt{P^2 + Q^2 + 2PQ \cos \theta}$$

$$\Rightarrow \sqrt{400 + 225 + 2 \times 20 \times 15 \times \cos 60^\circ}$$

$$\Rightarrow \sqrt{625 + 2 \times 20 \times 15 \times \frac{1}{2}}$$

$$\Rightarrow \sqrt{625 + 300} \Rightarrow \sqrt{925}$$

$$R \Rightarrow 30.41$$



$$\tan \alpha \Rightarrow \frac{Q \sin \theta}{P + Q \cos \theta} \Rightarrow \frac{15 \sin 60^\circ}{20 + 15 \cos 60^\circ}$$

$$\Rightarrow \frac{15 \times \frac{\sqrt{3}}{2}}{20 + 15 \times \frac{1}{2}}$$

$$\Rightarrow \frac{25.98}{2}$$

$$\frac{55.96}{2}$$

$$\Rightarrow \frac{25.98}{55} \Rightarrow 0.4723$$

$$\alpha \Rightarrow \tan^{-1}(0.4723)$$

$$\Rightarrow 25.28$$

Case II:-

$$R \Rightarrow \sqrt{625 + 2 \times 20 \times 15 \cos 120^\circ}$$

$$R \Rightarrow \sqrt{625 + 2 \times 20 \times 15 \left(-\frac{1}{2}\right)}$$

$$R \Rightarrow \sqrt{625 - 300} \Rightarrow \sqrt{325}$$

$$R \Rightarrow 18.02$$

$$\tan \alpha = \frac{Q \sin(120^\circ)}{P + Q \cos 120^\circ} \Rightarrow \frac{15 \times 0.866}{20 + 15 \times (-0.5)}$$

$$\Rightarrow \frac{12.99}{20 - 7.5}$$

$$\Rightarrow \frac{12.99}{12.5}$$

$$\Rightarrow 1.0392$$

$$\alpha = \tan^{-1}(1.0392)$$

Q The resultant of two forces P & 30 N is 40 N inclined at 60° to the 30 N force. Find the magnitude & direction of P .

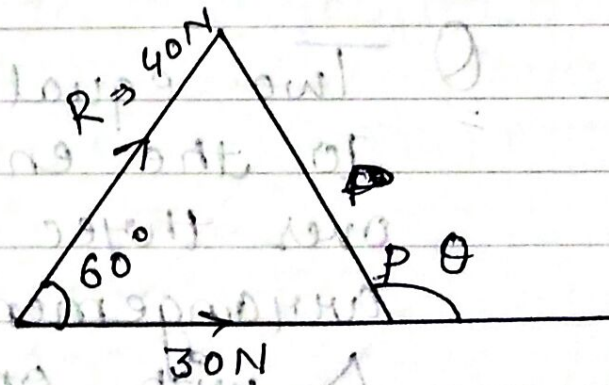
Sol. $R^2 = P^2 + Q^2 + 2PQ \cos \theta \Rightarrow 60$

$$(40)^2 = P^2 + (30)^2 + \frac{2 \times P \times 30 \times 1}{2}$$

$$1600 = P^2 + 900 + 30P$$

$$P^2 + 30P - 700 = 0$$

$$P = 36.05$$



$$\frac{P}{\sin 60^\circ} \Rightarrow \frac{R}{\sin(180-\theta)}$$

$$\boxed{\begin{array}{l} \sin(180-\theta) \\ \Rightarrow \sin \theta \end{array}}$$

$$\frac{36.05 \times 2}{\sqrt{3}} \Rightarrow \frac{R}{\sin \theta}$$

$$\sin \theta \Rightarrow \frac{\sqrt{3} R}{36.05 \times 2} \Rightarrow \frac{\sqrt{3} \times 40}{36.05 \times 2}$$

$$\sin \theta \Rightarrow \frac{69.28}{72.1} \Rightarrow 0.9608$$

$$\boxed{\theta \Rightarrow 73.9^\circ}$$

Q Two equal weights of 10N are attached to the end of a thin string it passes over three smooth pegs in a wall arrangement in the form of equilateral Δ with one side horizontal. Find the pressure of Pegs.

Soln

$$R_B \Rightarrow \sqrt{200 + 2 \times 100 \times (-0.866)}$$

$$\Rightarrow \sqrt{200 + (-173.2)}$$

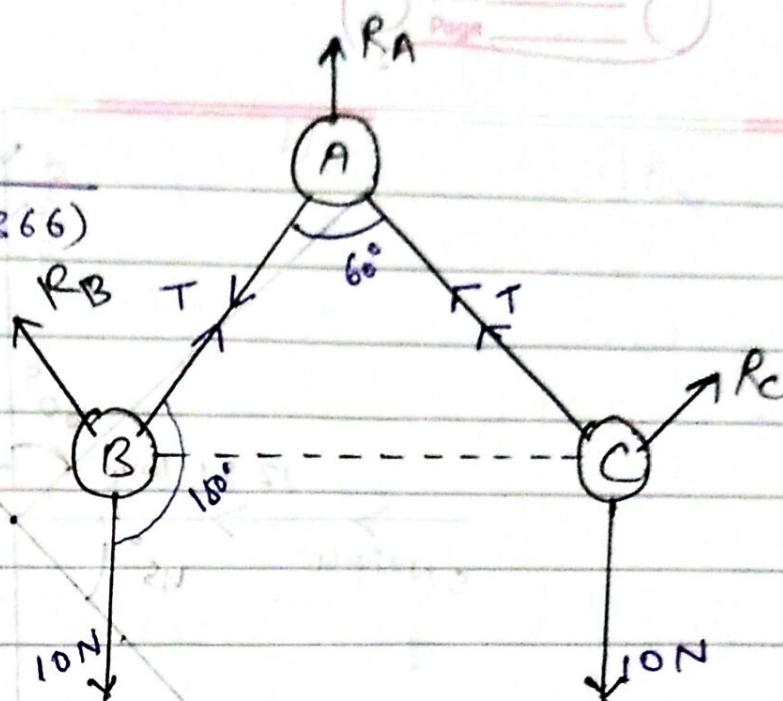
$$\Rightarrow \sqrt{26.8}$$

$$R_B \Rightarrow 5.17 \text{ N}$$

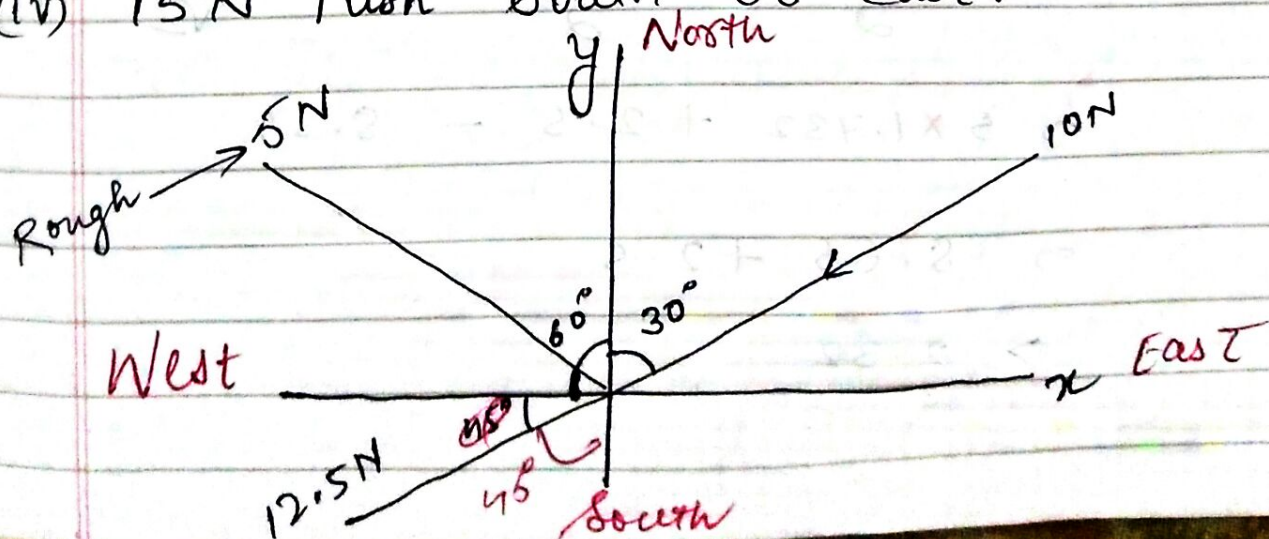
$$R_A \Rightarrow \sqrt{200 + 100}$$

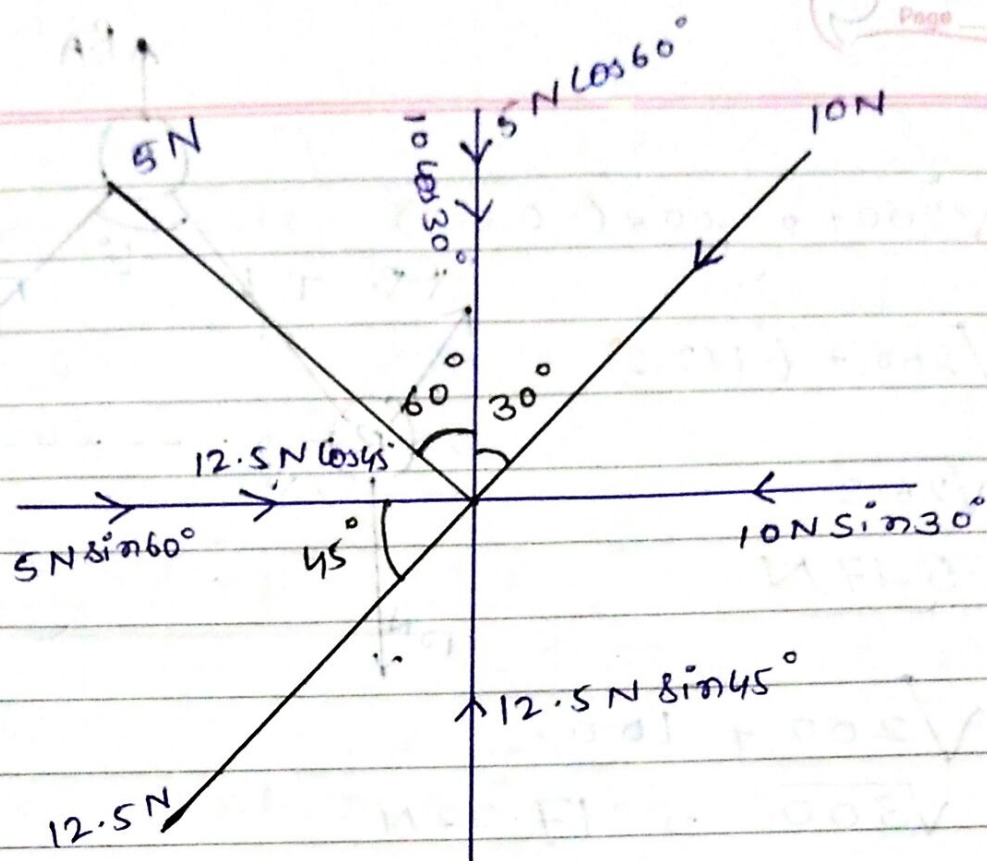
$$\Rightarrow \sqrt{300} \Rightarrow 17.32 \text{ N}$$

$$R_C \Rightarrow 5.17 \text{ N} \Rightarrow R_B$$



- ~~Q~~ Determine the Magnitude & direction of the resultant of the following four forces acting at a Point
- 10 N Pull north 30° east
 - 12.5 N Push south 45° West.
 - 5 N Push north 60° West.
 - 15 N Push south 60° East.



Sol.

$$\Sigma f_x \Rightarrow 12.5 \cos 45^\circ + 5 \sin 60^\circ - 10 \sin 30^\circ$$

$$\Rightarrow 12.5 \times \frac{1}{\sqrt{2}} + 5 \times \frac{\sqrt{3}}{2} - 10 \times \frac{1}{2}$$

$$\Rightarrow 8.84 + 4.33 - 5 \Rightarrow 8.17$$

$$\Sigma f_y \Rightarrow 10 \cos 30^\circ + 5 \cos 60^\circ - 12.5 \sin 45^\circ$$

$$\Rightarrow 10 \times \frac{\sqrt{3}}{2} + 5 \times \frac{1}{2} - 12.5 \times \frac{1}{\sqrt{2}}$$

$$\Rightarrow 5 \times 1.732 + 2.5 - 8.84$$

$$\Rightarrow 8.66 + 2.5$$

$$\Rightarrow 2.32$$

$$R \Rightarrow \sqrt{(8.17)^2 + (2.32)^2}$$

$$\Rightarrow \sqrt{66.74 + 5.38}$$

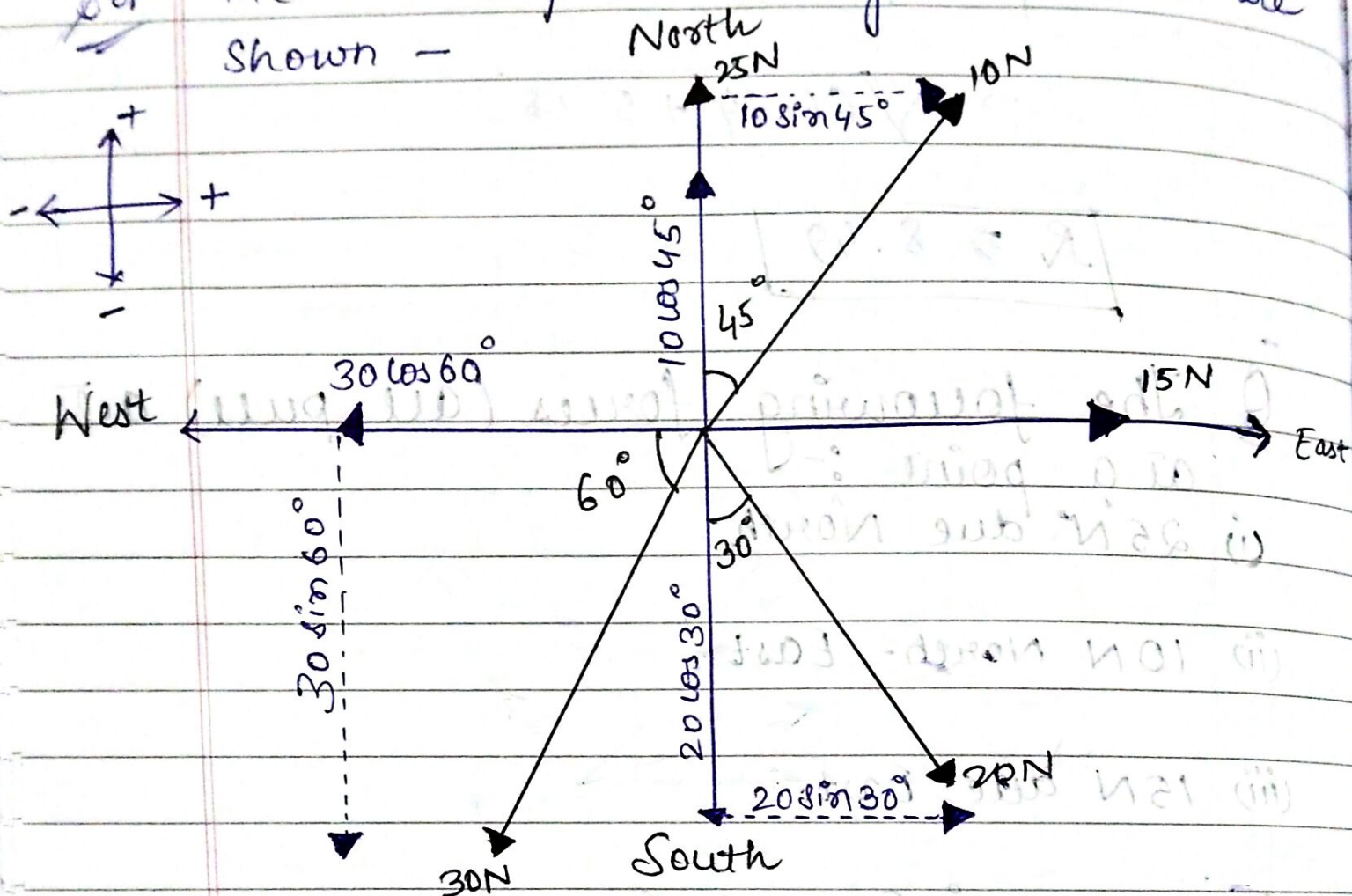
$$R \Rightarrow 8.49$$

Q The following forces (all pull) are at a point :-

- (i) 25 N due North.
- (ii) 10 N North-East.
- (iii) 15 N due East.
- (iv) 20 N 30° East of South.
- (v) 30 N 60° South of West.

Find the resultant force. What angle does it make with East?

Sol. The various forces acting at a Point are shown -



Resolving the forces horizontally,

$$\sum H \Rightarrow 10 \sin 45^\circ - 30 \cos 60^\circ + 20 \sin 30^\circ + 15$$

$$\Rightarrow 10 \times 0.707 - 30 \times 0.5 + 20 \times 0.5 + 15$$

$$\Rightarrow 7.07 - 15 + 10 + 15$$

$$\Rightarrow 17.07 \text{ N}$$

Resolving the forces Vertically,

$$\Sigma v \Rightarrow 10 \cos 45^\circ + 25 - 30 \sin 60^\circ - 20 \cos 30^\circ$$

$$\Rightarrow 10 \times 0.707 + 25 - 30 \times 0.866 - 20 \times 0.866$$

$$\Rightarrow 7.07 + 25 - 25.98 - 17.32$$

$$\Rightarrow -11.23 \text{ N}$$

Resultant, $R = \sqrt{(\Sigma H)^2 + (\Sigma v)^2}$

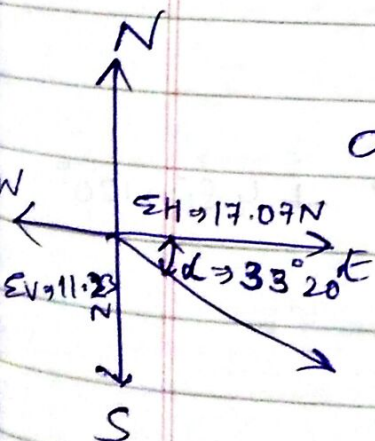
$$\Rightarrow \sqrt{(17.07)^2 + (-11.23)^2}$$

$$\Rightarrow \sqrt{291.38 + 126.11}$$

$$R \Rightarrow 20.43 \text{ N}$$

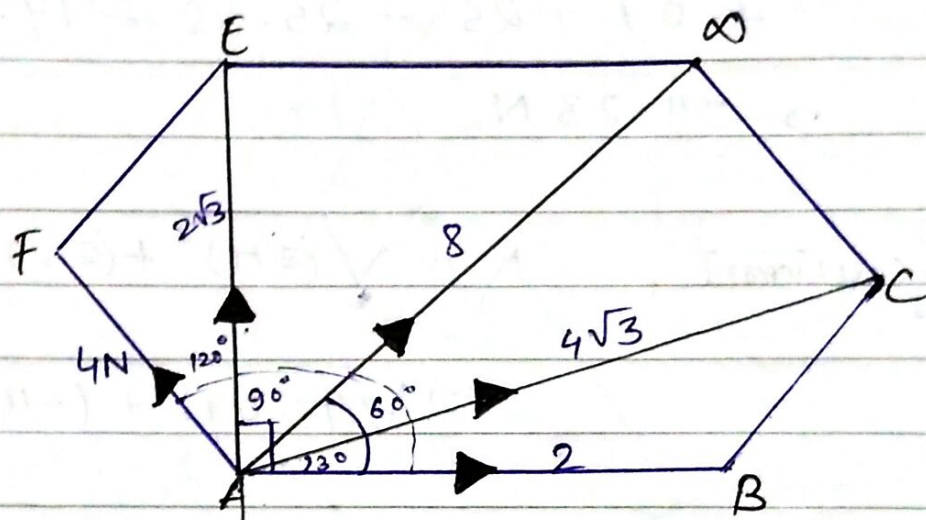
$$\tan \alpha \Rightarrow \left| \frac{\Sigma v}{\Sigma H} \right| \Rightarrow \left| \frac{-11.23}{17.07} \right| \Rightarrow \boxed{0.6578}$$

$\alpha \Rightarrow 33^\circ 20'$ South of East



Ex. ABCDEF is a regular hexagon. Forces of magnitude 2, $4\sqrt{3}$, 8, $2\sqrt{3}$ and 4N act at A in the directions of AB, AC, AD, AE and AF respectively. Determine the resultant completely.

Sol.



Forces along AB

$$\Rightarrow 2 + 4\sqrt{3} \cos 30^\circ + 8 \cos 60^\circ + 4 \cos 120^\circ$$

$$\Rightarrow \cancel{2} + 6 + 4 - \cancel{2} \Rightarrow 10 \text{ N}$$

Forces along AE

$$\Rightarrow 2\sqrt{3} + 4\sqrt{3} \sin 30^\circ + 8 \sin 60^\circ + 4 \sin 120^\circ$$

$$\Rightarrow 2\sqrt{3} + 2\sqrt{3} + 4\sqrt{3} + 2\sqrt{3}$$

$$\Rightarrow 10\sqrt{3} \text{ N}$$

$$\text{Resultant} \Rightarrow \sqrt{(10)^2 + (10\sqrt{3})^2}$$

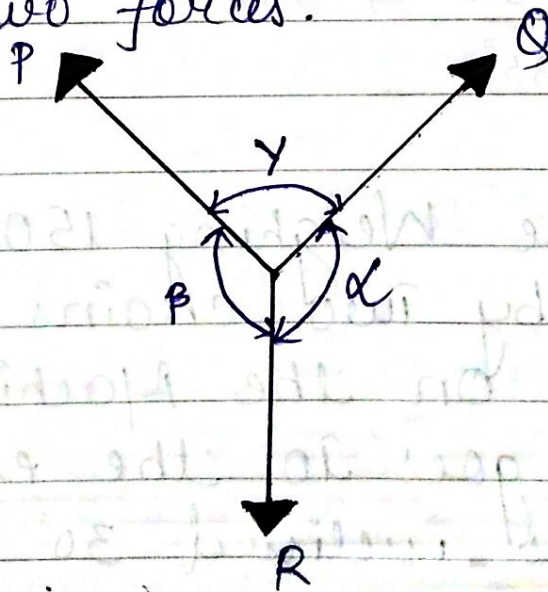
$$\Rightarrow \sqrt{100 + 300} \Rightarrow 20 \text{ N}$$

$$\tan \alpha \Rightarrow \frac{10\sqrt{3}}{10} \Rightarrow \sqrt{3} \Rightarrow \tan 60^\circ$$

$$\boxed{\alpha \Rightarrow 60^\circ}$$

LAMI'S THEOREM

It states that, "If three coplanar forces acting on a point in a body keep it in equilibrium, then each force is Proportional to the Sine of the angle between the other two forces."



$$\boxed{\frac{P}{\sin \alpha} \Rightarrow \frac{Q}{\sin \beta} \Rightarrow \frac{R}{\sin \gamma}}$$

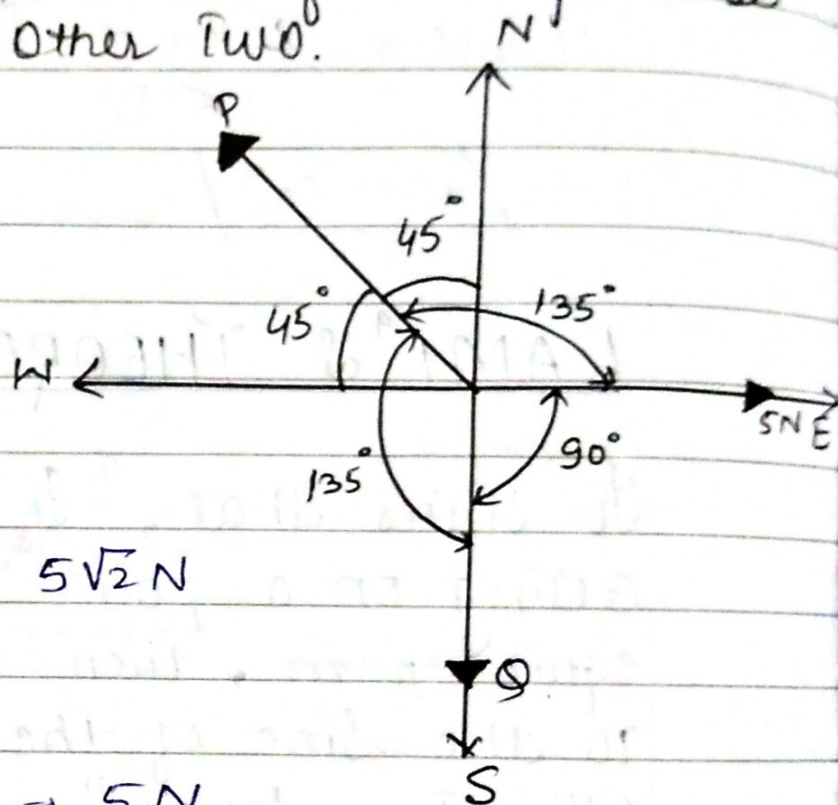
Q Three forces keep a Particle in equilibrium. One acts Towards east, Another Towards north-West and the third Towards south. If the first be 5N, find the other two.

Sol.

$$\frac{P}{\sin 90^\circ} \Rightarrow \frac{Q}{\sin 135^\circ} \Rightarrow \frac{5}{\sin 135^\circ}$$

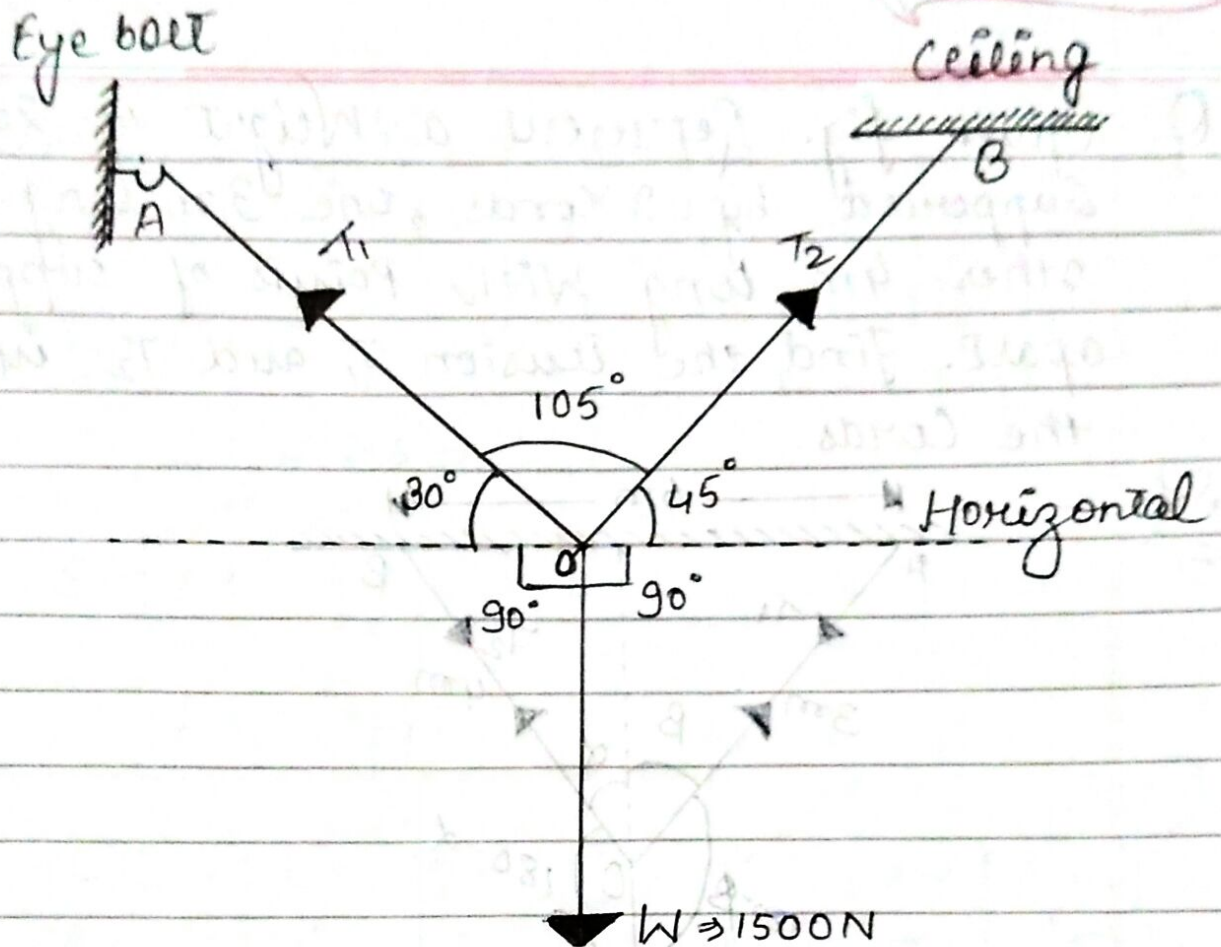
$$P \Rightarrow \frac{5 \sin 90^\circ}{\sin 135^\circ} \Rightarrow 5\sqrt{2} \text{ N}$$

$$Q \Rightarrow \frac{5 \sin 135^\circ}{\sin 135^\circ} \Rightarrow 5 \text{ N}$$



Q A Machine Weighing 1500N is supported by two chains attached to some point on the Machine. One of these Ropes goes to the eye bolts in the Wall and is inclined 30° to the horizontal and other goes to the hook in Ceiling and is inclined at 45° to the horizontal. Find the Tensions in the Two Chains.

Soln



Applying Lami's theorem at O,

$$\frac{T_1}{\sin(90^\circ + 45^\circ)} = \frac{T_2}{\sin(90^\circ + 30^\circ)} = \frac{W = 1500}{\sin 105^\circ}$$

$$\begin{aligned} \sin 105^\circ &= \sin(60^\circ + 45^\circ) \\ &= \sin 60^\circ \cos 45^\circ + \cos 60^\circ \sin 45^\circ \end{aligned}$$

$$T_1 = \frac{1500 \times \sin 135^\circ}{\sin 105^\circ} = \frac{1500 \times 0.707}{0.965}$$

$$\Rightarrow 1098.96 \text{ N } \underline{\underline{\text{Ans}}}$$

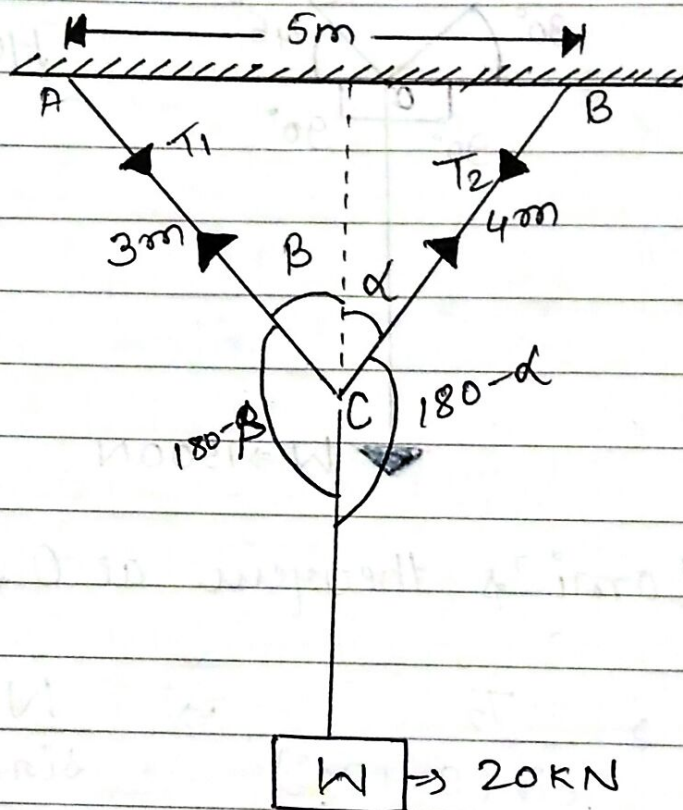
$$T_2 = \frac{1500 \times \sin 120^\circ}{\sin 105^\circ} = \frac{1500 \times 0.866}{0.965}$$

$$\Rightarrow 1346.11 \text{ N } \underline{\underline{\text{Ans}}}$$

$$\begin{aligned} &\frac{\sqrt{3}+1}{2\sqrt{2}} \\ &\approx \frac{2.732}{2.828} \\ &\Rightarrow 0.966 \end{aligned}$$

Q Given fig. Represent a Weight of 20 kg N Supported by 2 Cords, one 3 m long and the other 4 m long With Points of support 5 m apart. Find the Tension T_1 and T_2 in KN in the Cords.

Soln



Soln

$$AB^2 \Rightarrow AC^2 + BC^2$$

$$(5)^2 \Rightarrow (3)^2 + (4)^2$$

$$[25 \Rightarrow 25]$$

Also by sine rule,

$$\frac{3}{\sin \angle B} \Rightarrow \frac{4}{\sin \angle A} \Rightarrow \frac{5}{\sin \angle C}$$

$$\frac{3}{\sin LB} = \frac{4}{\sin LA} = \frac{5}{\sin 90^\circ}$$

$$\sin LB = \frac{3}{5} = 0.6$$

$$LB = 36^\circ 52'$$

$$\sin LA = \frac{4}{5} = 0.8$$

$$LA = 53^\circ 8'$$

$$L\alpha = 90^\circ - LB = 90 - 36^\circ 52' = 53^\circ 8'$$

$$L\beta = 90^\circ - LA = 90 - 53^\circ 8' = 36^\circ 52'$$

$$\frac{T_1}{\sin(180-\alpha)} = \frac{T_2}{\sin(180-\beta)} = \frac{W}{\sin(\alpha+\beta)}$$

$$\frac{T_1}{\sin \alpha} = \frac{T_2}{\sin \beta} = \frac{W}{\sin(\alpha+\beta)}$$

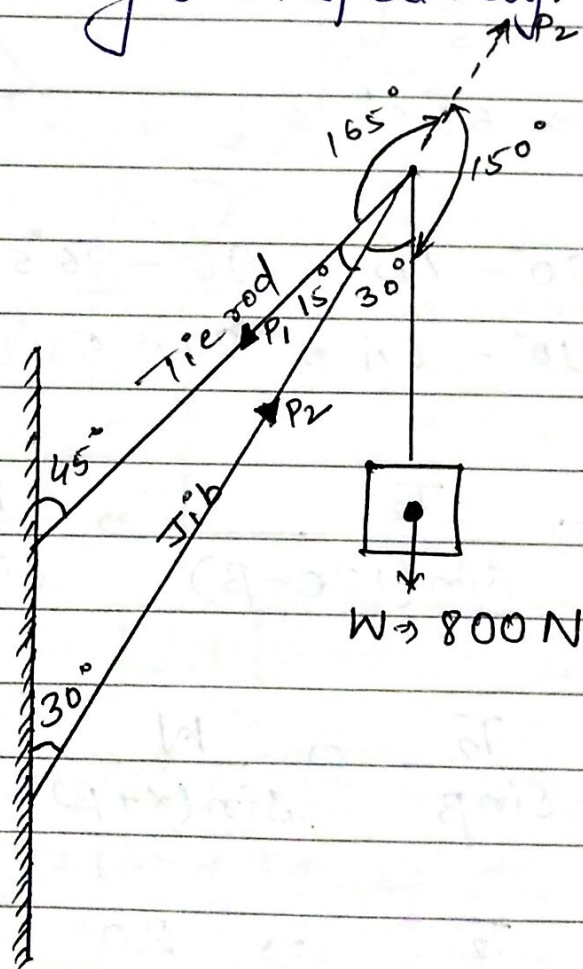
$$\frac{T_1}{\sin 53^\circ 8'} = \frac{T_2}{\sin 36^\circ 52'} = \frac{20}{\sin 90^\circ}$$

$$T_1 = \frac{20}{\sin 90^\circ} \times \sin 53^\circ 8' = 16 \text{ kN}$$

$$T_2 = \frac{20}{\sin 90^\circ} \times \sin 36^\circ 52' = 12 \text{ kN}$$

Q what axial forces does the vertical load $W = 800\text{ N}$ induce in the tie rod and the jib of the jib crane shown in fig. Neglect the self-weight of the members.

Sol. let P_1 and P_2 be the forces induced in the tie rod and jib respectively.



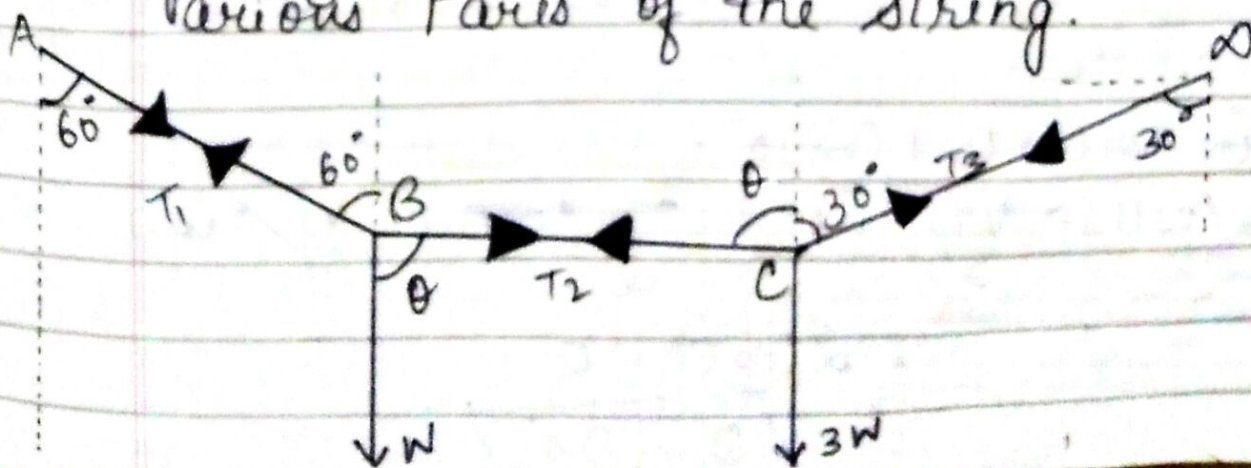
$$\frac{P_1}{\sin(180^\circ - 30^\circ)} = \frac{P_2}{\sin(30^\circ + 15^\circ)} = \frac{W}{\sin(180^\circ - 15^\circ)}$$

$$\frac{P_1}{\sin 30^\circ} \Rightarrow \frac{P_2}{\sin 45^\circ} \Rightarrow \frac{800}{\sin 15^\circ}$$

$$P_1 \Rightarrow \frac{800 \sin 30^\circ}{\sin 15^\circ} \Rightarrow \frac{800 \times 0.5}{0.2588} \\ \Rightarrow 1545.6 \text{ N}$$

$$P_2 \Rightarrow \frac{800 \sin 45^\circ}{\sin 15^\circ} \Rightarrow \frac{800 \times 0.707}{0.2588}$$

Q The extremities A and D of a light inextensible string ABCD are tied to two points in the same horizontal line. Weight W and 3W are tied to two points in the same horizontal line the string at the points B and C respectively. If AB and CD are inclined to the vertical at angles 60° and 30° respectively, show that BC is horizontal and find the tensions in the various parts of the string.



Applying Lami's Theorem, at pt B.

$$\frac{T_1}{\sin \theta} \Rightarrow \frac{T_2}{\sin 120^\circ} \Rightarrow \frac{W}{\sin (240^\circ - \theta)} \quad \text{--- (1)}$$

Sly. Applying Lami's Theorem at pt C,

$$\frac{T_2}{\sin 150^\circ} \Rightarrow \frac{T_3}{\sin (180^\circ - \theta)} \Rightarrow \frac{3W}{\sin (30^\circ + \theta)} \quad \text{--- (2)}$$

from eq - (1) & (2)

$$T_2 \Rightarrow \frac{W \sin 120^\circ}{\sin (240^\circ - \theta)} \Rightarrow \frac{3W \sin 150^\circ}{\sin (30^\circ + \theta)}$$

$$\frac{\sin 60^\circ}{-\sin (60^\circ - \theta)} \Rightarrow \frac{3 \sin 30^\circ}{\sin (30^\circ + \theta)}$$

$$\Rightarrow \frac{\sqrt{3}}{2} \sin (30^\circ + \theta) \Rightarrow -3 \times \frac{1}{2} \sin (60^\circ - \theta)$$

$$\Rightarrow \sin 30^\circ \cos \theta + \cos 30^\circ \sin \theta$$

$$\Rightarrow -\sqrt{3} (\sin 60^\circ \cos \theta - \cos 60^\circ \sin \theta)$$

$$\Rightarrow \sin 30^\circ \cos \theta + \cos 30^\circ \sin \theta$$

$$\Rightarrow \frac{1}{2} \cos \theta + \frac{\sqrt{3}}{2} \sin \theta \Rightarrow -\frac{3}{2} \cos \theta + \frac{\sqrt{3}}{2} \sin \theta$$

$$2 \cos \theta = 0$$

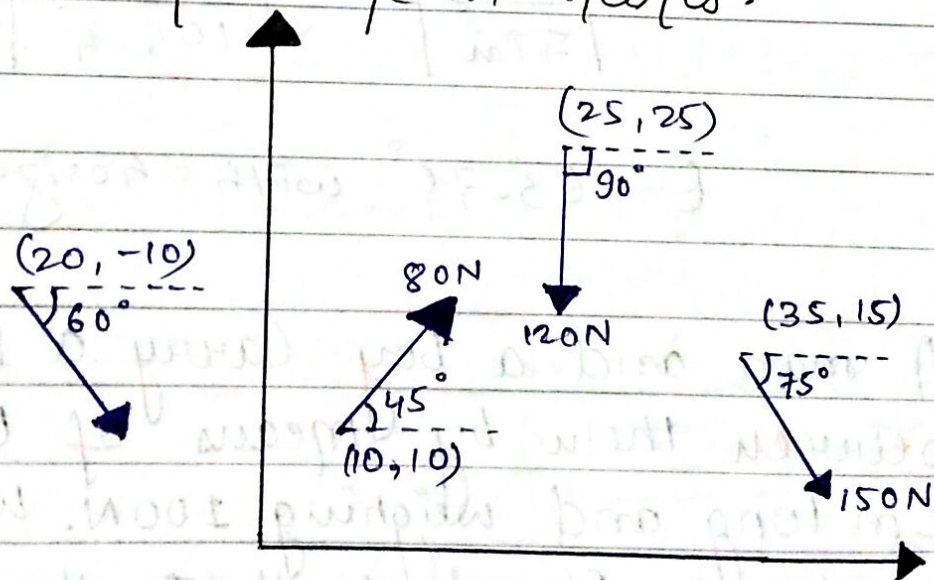
$$\theta = 90^\circ$$

$$T_1 \Rightarrow \frac{W \sin 90^\circ}{W \sin 150^\circ} \Rightarrow \frac{W}{\frac{1}{2}} \Rightarrow 2W.$$

$$T_2 \Rightarrow \frac{W \sin 120^\circ}{\sin 150^\circ} \Rightarrow \frac{\frac{\sqrt{3}}{2} W}{\frac{1}{2}} \Rightarrow \sqrt{3} W.$$

$$T_3 \Rightarrow \frac{3W \sin 90^\circ}{\sin 120^\circ} \Rightarrow \frac{3W}{\frac{\sqrt{3}}{2}} \Rightarrow 2\sqrt{3} W.$$

Q. Determine the resultant of the force acting system shown in fig. Assume that the coordinates are in metres.



$$\begin{aligned} \sum P_x &\Rightarrow 100 \cos 60^\circ + 80 \cos 45^\circ + 150 \cos 75^\circ \\ &\Rightarrow 100 \times 0.5 + 80 \times 0.707 + 150 \times 0.259 \\ &\Rightarrow 145.4 \text{ N} \end{aligned}$$

$$\begin{aligned}\Sigma P_y &\Rightarrow -100 \sin 60^\circ + 80 \sin 45^\circ - 120 - 150 \sin 75^\circ \\ &\Rightarrow -100 \times 0.866 + 80 \times 0.707 - 120 - 150 \times 0.966\end{aligned}$$

$$\Rightarrow -86.6 + 56.56 - 120 - 144.9$$

$$\Rightarrow -294.94 \text{ N}$$

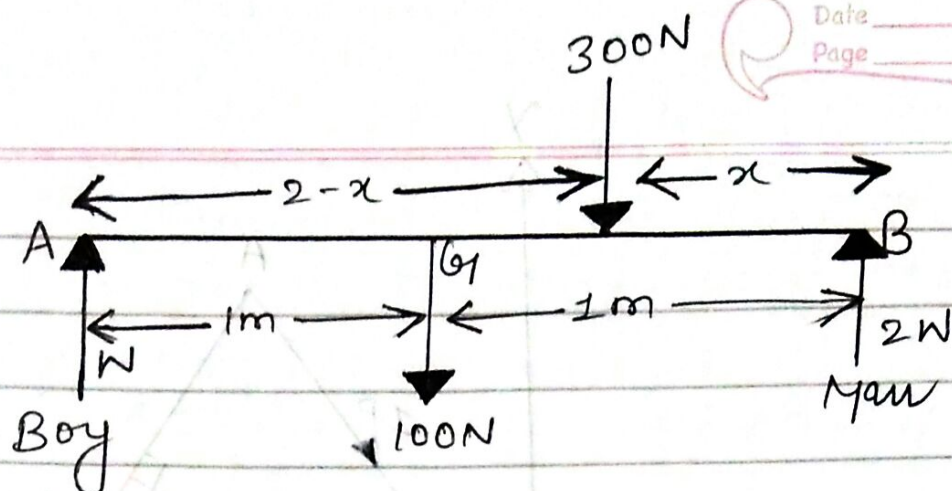
$$R \Rightarrow \sqrt{(\Sigma P_x)^2 + (\Sigma P_y)^2}$$

$$\Rightarrow \sqrt{(145.4)^2 + (-294.94)^2} \Rightarrow \boxed{328.83 \text{ N}}$$

$$\tan \theta = \frac{|\Sigma P_y|}{|\Sigma P_x|} \Rightarrow \frac{294.94}{145.4} \Rightarrow 2.03$$

$$\theta = 63.75^\circ \text{ with horizontal}$$

Q A man and a boy carry a weight of 300N between them by means of uniform pole 2m long and weighing 100N. Where must the weight be placed so that the man may bear twice as much of the weight as that of the boy?

Solⁿ

$$\sum V \Rightarrow 0$$

$$W + 2W \Rightarrow 300 + 100 \Rightarrow 400$$

$$W \Rightarrow \frac{400}{3} \Rightarrow 133.3 \text{ N}$$

Taking Moments about A (boy)

$$2W \times 2 \Rightarrow 300(2-x) + 100 \times 1$$

$$4W \Rightarrow 600 - 300x + 100$$

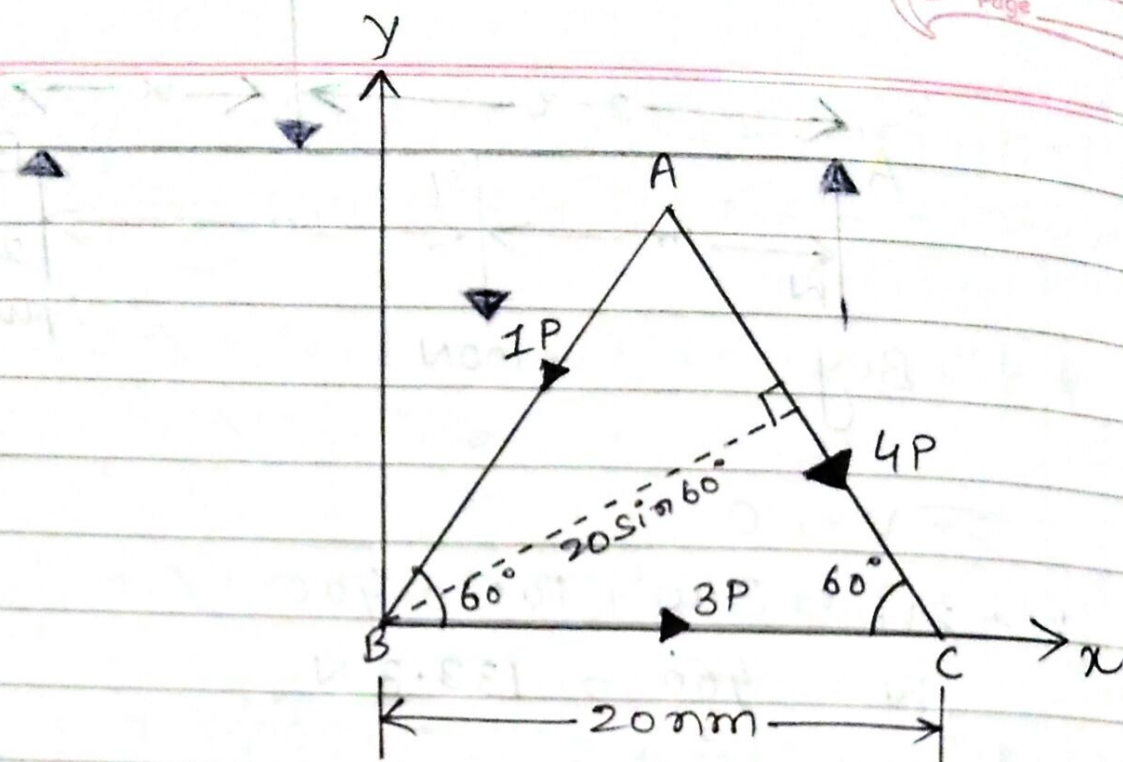
$$4 \times 133.3 \Rightarrow 700 - 300x$$

$$533.2 \Rightarrow 700 - 300x$$

$$300x \Rightarrow 166.8$$

$$x \Rightarrow 0.556 \text{ m}$$

Q forces $1P$, $3P$, $-4P$ act in order round the sides of an equilateral Triangle of side 20mm drawn on a rigid lamina. Find the Magnitude, direction and Position of their Resultant (line of action).

Soln

$$\begin{aligned}\Sigma H &\Rightarrow -1P \cos 60^\circ + 3P + 4P \cos 60^\circ \\ &\Rightarrow -0.5P + 3P + 2P = 4.5P\end{aligned}$$

$$\Sigma V \Rightarrow -1P \sin 60^\circ - 4P \sin 60^\circ \Rightarrow -4.33P$$

$$R \Rightarrow \sqrt{(\Sigma H)^2 + (\Sigma V)^2}$$

$$\Rightarrow \sqrt{(4.5P)^2 + (-4.33P)^2}$$

$$\boxed{R \Rightarrow 6.24P}$$

$$\tan \theta \Rightarrow \frac{\Sigma V}{\Sigma H} \Rightarrow \frac{-4.33P}{4.5P} \Rightarrow -0.96$$

$$\theta \Rightarrow -43.9^\circ \text{ (with horizontal)}$$

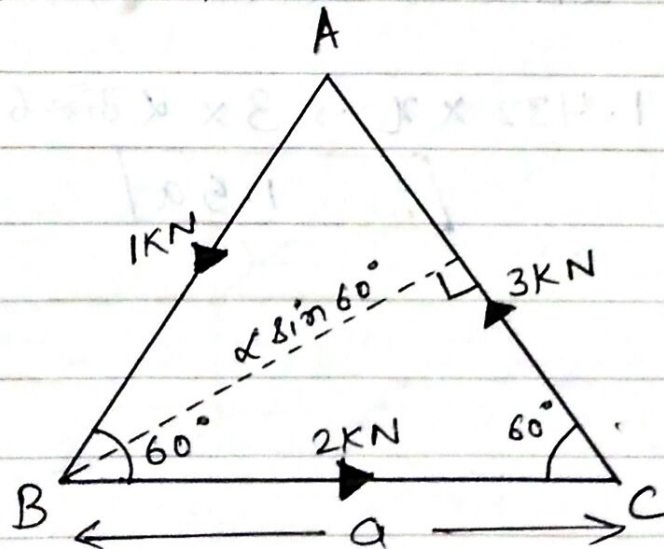
Let $x \Rightarrow$ Perpendicular distance between B and the line of the resultant force.

$$\sum M_B \Rightarrow 6.24P \times x \Rightarrow \cancel{6.24}P \times 0 + 3P \times 0 + 4P \times 20 \sin 60^\circ$$

$$x \Rightarrow 11.1 \text{ mm}$$

Q forces 1 kN, 2 kN and 3 kN act along the sides AB, BC and CA of a given eq. Triangle ABC. Find the magnitude and direction of their resultant, and find the point in which its line of action meets the side BC.

Solⁿ



$$\sum H \Rightarrow -1 \cos 60^\circ + 2 - 3 \cos 60^\circ$$

$$\Rightarrow -0.5 + 2 - 1.5$$

$$\Rightarrow 0$$

$$\Sigma V \Rightarrow -1 \sin 60^\circ + 3 \sin 60^\circ$$

$$\Rightarrow 2 \sin 60^\circ$$

$$\Rightarrow \frac{2 \times \sqrt{3}}{2} \Rightarrow 1.732 \text{ kN} \uparrow$$

$$R \Rightarrow \sqrt{(\Sigma H)^2 + (\Sigma V)^2}$$

$$\Rightarrow \sqrt{0 + (1.732)^2} \Rightarrow 1.732 \text{ kN} \uparrow$$

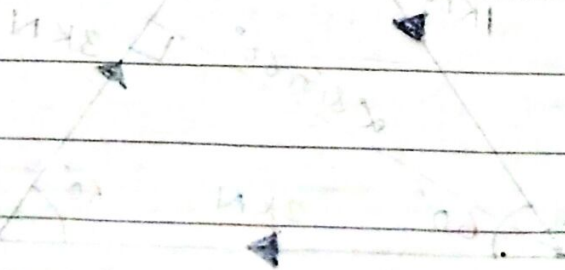
$$\tan \theta \Rightarrow \frac{\Sigma V}{\Sigma H} \Rightarrow \frac{1.732}{0} \Rightarrow \infty$$

$$\boxed{\theta \Rightarrow 90^\circ}$$

Taking moments about B,

$$1.732 \times x \Rightarrow 3 \times x \sin 60^\circ$$

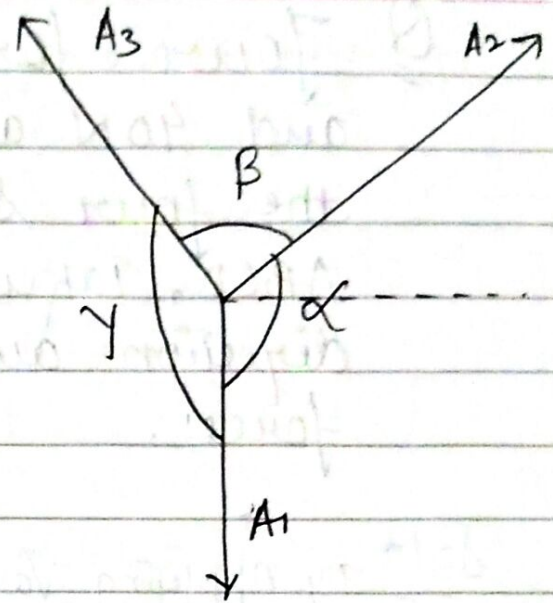
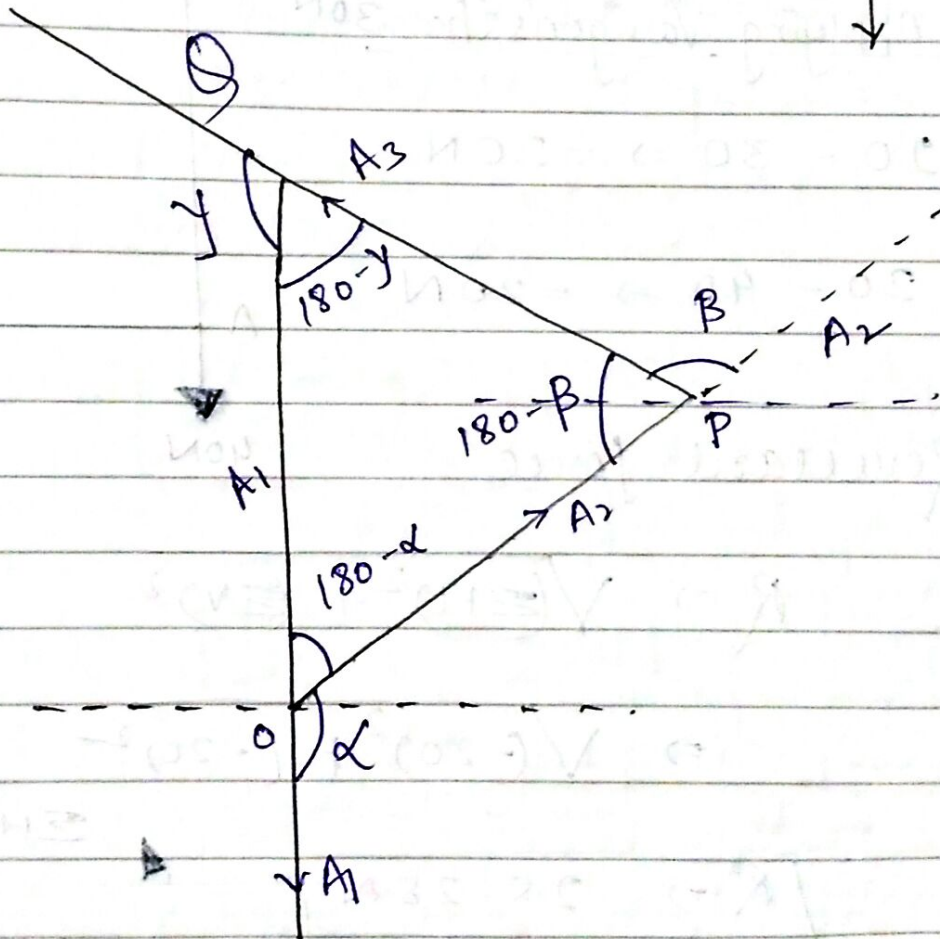
$$\boxed{x \Rightarrow 1.5a}$$



LAMI'S THEOREMLAMI'S THEOREM

By using Sine rule -

$$\frac{A_3}{\sin \alpha} \Rightarrow \frac{A_1}{\sin \beta} \Rightarrow \frac{A_2}{\sin \gamma}$$

Proof

$$\frac{A_1}{\sin(180-\beta)} \Rightarrow \frac{A_2}{\sin(180-\gamma)} \Rightarrow \frac{A_3}{\sin(180-\alpha)}$$

$$\frac{A_1}{\sin \beta} \Rightarrow \frac{A_2}{\sin \gamma} \Rightarrow \frac{A_3}{\sin \alpha}$$

Q Four forces equal to 10N, 20N, 30N and 40N are respectively acting along the four sides (1m each) of a square ABCD, taken in Order. Find the magnitude, direction and position of the resultant forces.

Soln By Applying Varignon's theorem

$$\Sigma H \Rightarrow 10 - 30 \Rightarrow -20N$$

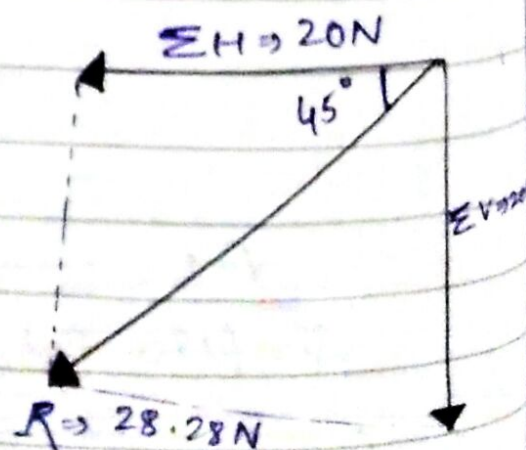
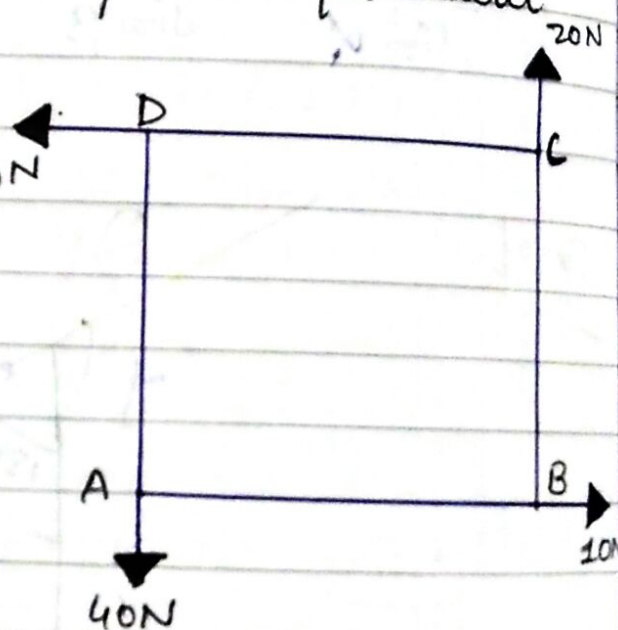
$$\Sigma V \Rightarrow 20 - 40 \Rightarrow -20N$$

Now, Resultant force

$$R \Rightarrow \sqrt{(\Sigma H)^2 + (\Sigma V)^2}$$

$$\Rightarrow \sqrt{(-20)^2 + (-20)^2}$$

$$R \Rightarrow 28.28N$$



Let $\theta \Rightarrow$ angle which the resultant makes with the horizontal.

$$\tan \theta \Rightarrow \frac{\sum V}{\sum H} \Rightarrow \frac{-20}{-20} \Rightarrow 1$$

$$\boxed{\theta \Rightarrow 45^\circ}$$

Since $\sum H$ as well as $\sum V$ is $-ve$, therefore θ lies between 180° and 270° .

Actual, $\theta \Rightarrow 180^\circ + 45^\circ \Rightarrow 225^\circ$.

Let $x \Rightarrow$ Perpendicular Distance between A and the line of the resultant forces.

Now Taking Moments about A, we get

$$\sum M_A \Rightarrow 28.28 \times x \Rightarrow 10 \times 0 + 20 \times 1 + 30 \times 1 + 40 \times 0$$

$$\Rightarrow 20 + 30$$

$$\Rightarrow 50$$

$$x \Rightarrow \frac{50}{28.28} \Rightarrow \boxed{1.768 \text{ m}}$$